

Exercise/Activity Plan

Regular lifelong exercise/activity

Keeping it going

- Remember it is important to keep building up your walking or any new activity that you have taken up.
- Do not make more than one change at a time.** This can relate to any new activity you start up. Think about changing the distance you go or the length of time you do the activity, the speed you do the activity or the amount of effort you put in.
- Remember you are aiming to breathe faster and feel a little warmer.
- Remember this needs to be an activity you can do every day.
- Keep filling in your walking record as before, and if you are doing a new activity add it in as well, and build up as your activity becomes easier.

Make it your routine

- Prioritise activity time
- Make it part of your daily routine such as walking to get the papers or go for a walk in your lunch break with friends
- Find an activity you enjoy
- Join a club or a class
- Go with a friend

Activity needs to become part of your lifestyle if you want to build and maintain the good start you have made while working on the Heart Manual. Some leisure centres have staff with specialist training and can help people to safely build up their exercise after cardiac rehabilitation. Speak to your facilitator about referral or what is available in your area.

Set Back Plan

Set backs are a normal part of life but they can interfere with following the Heart Manual Plan. These could be going on holiday, moving house, visitors or feeling unwell. What would you do? Give up all the good work you have done up till now? Hopefully not.

If it was due to holidays, moving house or visitors try and maintain as much as you can. Bear in mind you may not be doing your normal exercise/activity plan but you may be active in other ways. Once things have settled down re-write your plan and prioritise your exercise/activity. Build up gradually until you are back to where you were prior to the break.

If you are unwell stop or cut back on your walking and exercises. When you are feeling better remember to set your targets at a level which is easy and build up again. **If you have been very unwell or are unsure about returning to your exercise/activity plan speak to your doctor or facilitator.**

Walking record

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard
Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard
Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard
Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard
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Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard
Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Date	How long exercised	How many times		
That was				
	Too Easy	Fairly Easy	Fairly Hard	Too Hard

Write down the things you do and how you feel each day

Date

Date

Date

Date

Date

Date

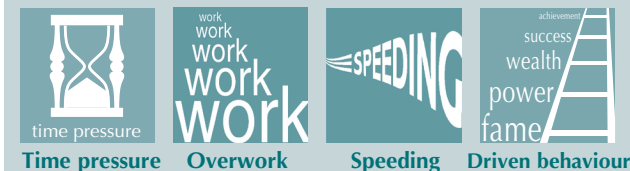
Date

Date

Do you ever get so angry you could explode?

The story so far ...

Over the past weeks we have looked at several different kinds of stress:



We have seen how stress affects us by raising our adrenaline level. Adrenaline is a chemical produced by the body.

At normal levels, or in emergencies, adrenaline is very useful (if you want to know how it helps in emergencies, look at pages 156-166 in Part 3 of this Manual). Adrenaline becomes a problem when we have high levels in our blood all the time. This makes us feel tired and sick, and affects our mood.

Stress alone isn't a major risk factor

At present there is some evidence to suggest work stress can have a small effect on later development of coronary artery disease.

BUT ...

BUT it's possible that people who tend to experience anger, annoyance or irritation more frequently or intensely than others may be more likely to build up the risk factors which lead to coronary artery disease. And it may be that these feelings can affect recovery. If you sometimes 'blow your top' or store up a lot of intense emotions inside, this section might help.

Hostility doesn't mean punching someone

Psychologists call these feelings **hostility**. Having hostility doesn't mean that you are going to become violent or hurt someone. It means that you have some strong emotional feelings.

The BAD NEWS about hostility

People with a lot of hostility can:

- mistrust and blame others when frustrated
- feel angry with those they blame
- behave aggressively.

Hostility can interfere with happy and healthy relationships. It can lead to social isolation. It can interfere with your ability to listen to and take advice from your loved ones and your healthcare providers.

People with high levels of hostility tend to have greater increases in adrenaline when they get stressed. And they have more difficulty in relaxing afterwards.

The GOOD NEWS

There are lots of things you can do to cope with hostility.

Coping effectively with hostility can:

- improve your physical health
- decrease your stress
- make you less irritable
- enhance your relationships with others
- improve the overall quality of your life.

Dealing with hostility and anger

You can learn to control hostility and anger. Here are some tips.

1 Learn to identify your triggers and reactions

Keep a diary of situations in which you feel angry, frustrated or annoyed.

2 Challenge your automatic thoughts

Identify your 'negative automatic thoughts' – such as when you immediately assume the worst or find fault or blame others. Then challenge these automatic thoughts by examining the facts that support them and the facts that don't support them.

This is what Mary Evans did. While driving to work, she was cut in on by another driver and became instantly enraged. 'I could kill that selfish, arrogant **', she thought (fill in the blanks yourself). Then she took a step back from that thought and challenged it by looking at the facts that might contradict her automatic thought:**

'How do I know that the other driver was being selfish or arrogant?' 'Maybe he made an innocent mistake and didn't see me.' 'Maybe he's in a rush or late for something important – like I am sometimes.' 'Maybe he's a new driver without much experience.'

3 Decide how important the situation is

- Do you have all the facts?
- Is it worth your time and effort to make a big deal of the situation?
- Can you do something to change or resolve the situation?

Mary decided that there was nothing she could do. The situation was not critical. She reminded herself how lucky she was to be feeling so much better after her bypass operation. In the grand scheme of things, being cut in on by another driver really didn't matter all that much.

4 Use behavioural strategies

When you feel unable to control the intensity of your frustration, annoyance or anger, take a 'time out.' Take a break. Remove yourself from the situation. Take a big breath, relax, clear your mind, and get the situation back into perspective.

Practise saying **No**. Turn back to pages 98–99 and re-read how Linda North reduced her stress levels by saying No and making her own decisions about what she would and wouldn't do.

5 Use relaxation techniques

Keep using the relaxation techniques you have been practising over the last 6 weeks. Stopping smoking and cutting down on caffeine, alcohol and sugar intake can also help. So can exercise.

6 Be positive

Distract yourself from upsetting thoughts, situations and feelings. Try to find a little humour in every situation.

This week's risk factor: High blood pressure

Why does high blood pressure matter?

Blood pressure is like cholesterol – we all have it, but some people have more of it than others. Doctors have found that people whose blood pressure is at the top end of the range have an above-average risk of getting coronary artery disease and other circulation problems. So blood pressure matters because it's a **risk factor**.



How do I know if I've got high blood pressure?

From measuring blood pressure in thousands of people for many years, doctors have been able to work out levels for 'normal' or 'high' blood pressure. Because your heart pumps in beats, two blood pressure measurements are taken – a peak figure when the heart beats, and a lower one when it relaxes. Your doctor or nurse will express this as two numbers such as 130 over 80. It is important that you have your blood pressure checked regularly so that you know what your blood pressure should be. Remember that these figures are for when you are resting, and it is normal for blood pressure to go up and down depending on your activity.

What causes high blood pressure?

For most people with high blood pressure there is probably no special cause, but there are things which tend to put the blood pressure up into the 'high' range.

These include **being overweight, lack of exercise, drinking too much alcohol, smoking** and **kidney disease**. **Too much salt** in your diet can also raise your blood pressure. Some doctors think that stress can raise it, too.

What can be done about it?

There's good news! There are many medicines that can lower blood pressure, and the risk of developing coronary artery disease decreases at the same time. And if you or your doctor thinks that your blood pressure is caused by any of the factors listed above, you can do your bit to help by:

- reducing your alcohol intake
- losing weight by dieting
- taking less salt in your food
- practising your relaxation – even if you don't feel stressed
- taking regular exercise
- stop smoking
- take your medication as prescribed
- attend GP surgery appointments for regular checks of your BP.

Back to work

This page is mainly for people who have a job. If you had an angioplasty (stent or balloon), you are probably back at work already. If you had a bypass, you may be wondering when it's safe to go back. If you have a non-manual occupation, such as office, administrative or managerial work, you can think about returning to work from about 2 months after your operation. If you have a heavy manual job you may not be able to return to work until at least 3 months after your operation, when your breast bone and muscles in your chest are fully healed. Returning to work can be an anxious time, but many people who return after a heart operation are doing their full range of duties within a few months.

Here are some tips which might help.

Get advice first

When you feel ready to go back to work, talk to your doctor or facilitator first. They can advise you whether it's safe to go back yet.

Contact your employer

Get in touch with your employer and discuss your plans for coming back. Most employers will be very understanding and will let you do shorter hours with lighter duties at first.

Build up your working hours gradually

It may not be sensible to go straight back to exactly the same routine as you had before your procedure, particularly if you have had a bypass operation. Plan to start with fewer working hours and gradually build up to your normal hours over an agreed number of weeks. For example, you could:

- take work home initially – this might be a good option if you are a desk worker, manager, etc.
- start with either mornings or afternoons, or
- spend alternate days working and not working
- then increase your hours gradually until you are back to normal.

Start with lighter duties

Make a list of all the tasks that you do in your job – daily, weekly, monthly and one-off jobs. Put them in order from the easiest to the most difficult.

Discuss and agree with your bosses that at first you will only do the lighter duties. Over an agreed timescale you can gradually get back to doing the harder ones.

Make sure that you get feedback

Ask your employer to name someone (like a manager or foreman) who will be able to give you regular feedback on how you are doing, and who can decide with you which extra tasks you should take on and when.

Take all your rest periods and breaks

Try not to miss rest periods and meal breaks. If the job will allow, it's a good idea to take one or two extra rests during the course of the day.

When you go back to work, try to remember some of the things you have read about in previous weeks of this programme. If you think your old ways of working may have been adding to your risk factors, then you will probably agree that it wouldn't be a good idea to let yourself go back to them. Don't forget that the Manual is here to help.

- Re-read the sections about **time pressure, deadlining, the Superhero complex, the Yes complex** and **stress at work** on pages 94–101.
- Re-read the sections about **speeding, overworking** and **driven behaviour** on pages 112–119.
- Try to get regular exercise during the working day.
- Remember to do your relaxation while you are at work.

Daily Tasks	
Order processing	1
Answering phone calls	1
Filing	2
Post office	4
Packing and despatch	5
Weekly Tasks	
Stock check/ordering	3
Local van deliveries	4
Unpacking stoves	5
One-off Tasks	
Sort cabinets	3
New stock control systems	4
Re-organise stove room	5

What has happened over the last 6 weeks

Over the last 6 weeks we have covered all of the most important things that you need to know about and to do after a heart procedure.

Information:

- what coronary artery disease is and what causes it
- wrong ideas about coronary artery disease
- how to fight back and reduce the chances of having further heart problems.

Exercise

You have been helping your recovery and improving the strength of your body by using the Exercise/Activity Plan.

Relaxation

By now you should know from the Manual and the relaxation CD/audio about all the different ways in which relaxation can help you.

You should be able to notice tension developing and have ways to reduce it. Make these a part of your life without the relaxation CD/audio.

Stress reduction

You have looked at some of the ways in which stress can affect you, and what you can do about it.

Risk factors

You have looked at the main controllable risk factors and how to control them. These risk factors are **smoking, poor diet, being overweight, lack of exercise, high blood pressure** and **stress**.

Other worries

You know that coronary artery disease doesn't mean that you have to give up the more enjoyable things in life, like alcohol, eating out and sex.

You may also have used Part 3 of this Manual to find out more about:

- medicines – what they do for you and their side-effects
- the after-effects of your procedure
- anxiety after a heart procedure – why it happens and how to cope
- stress in relationships
- angina, other chest pains and palpitations
- low spirits and how to reduce them.

Coping with setbacks

By now you may be feeling much better. You may be wondering what all the fuss was about. That's fine, so long as you don't forget about the things you need to do to protect yourself in the future.

Or you may still be feeling less than 100%. That's only to be expected, so long as you don't let it get you down.

There is no 'normal' time for recovering. Indeed we wouldn't expect some people to be fully fit for some months yet.

Week by week you will be getting stronger.

Sometimes you may feel as if you have stopped making progress. It's only when you look back a few months later that you can see that this isn't the case.

If it begins to worry you, look back over your activity or exercise sheets or discuss your feelings with your facilitator. See how much more you can do now compared with what you could do before.

What has happened over the last 6 weeks (continued)

Coping with setbacks (continued)

You may have had to deal with some breathlessness, infections, swelling and aches and pains. You may even have had to go back into hospital.

If so, there is no reason for despair. Everything in the Manual remains just as true.

Read through the true facts about coronary artery disease again to remind yourself.

If your recovery has not been as quick as you had hoped, this doesn't mean that the outlook is bleak. Far from it.

There are many things that can be done to cope with the complications that sometimes arise.

Finally, you may like to try this checklist

How many of these things are you doing to protect yourself?

I am reducing my risk factors by:

- ☐ stopping smoking
- ☐ reducing smoking
- ☐ reducing fat consumption
- ☐ losing weight
- ☐ eating less
- ☐ exercising more
- ☐ being more active (less sedentary)
- ☐ managing my stress levels

I am lowering my blood pressure by:

- ☐ relaxing
- ☐ losing weight
- ☐ reducing salt intake
- ☐ exercising more

I am eating more:

- ☐ fruit and vegetables
- ☐ fish
- ☐ wholemeal bread
- ☐ high-fibre cereal
- ☐ beans and lentils
- ☐ wholemeal rice and pasta

I am eating less:

- ☐ red meat
- ☐ bacon
- ☐ crisps, cakes and biscuits
- ☐ fried take-aways
- ☐ butter
- ☐ whole-fat cheese
- ☐ sugar
- ☐ and I'm drinking less alcohol

I am doing regular exercise:

- ☐ walking more
- ☐ cycling
- ☐ exercise classes
- ☐ other interesting and enjoyable exercise
- ☐ and I'm planning regular exercise or sport sessions in future

I relax regularly by:

- ☐ using the methods on the relaxation CD/audio
- ☐ using my own method
- ☐ using imagery
- ☐ using breathing exercises
- ☐ making regular times to relax
- ☐ finding new hobbies or interests

(continued overleaf)

What has happened over the last 6 weeks (continued)

I use rapid relaxation:

- ☐ when driving
- ☐ when angry, worried or fed up
- ☐ to manage pain

I work on reducing time pressure by:

- ☐ spending more time with family and friends
- ☐ making time for enjoyment
- ☐ using deadlining
- ☐ making regular time for myself
- ☐ saying No
- ☐ dropping the Superhero complex
- ☐ charging more for my time

I am reducing speeding by:

- ☐ getting control of my workload
- ☐ using relaxation
- ☐ using deadlining
- ☐ not putting things off
- ☐ taking things more slowly and steadily
- ☐ doing slowing-down exercises
- ☐ remaining calm when waiting

I am reducing overworking by:

- ☐ getting control of my workload
- ☐ increasing my other interests

- ☐ using relaxation
- ☐ changing my attitude to enjoying myself
- ☐ sorting out home tensions
- ☐ reducing my needs
- ☐ getting out of the habit

I am reducing driven behaviour by:

- ☐ changing my beliefs about success
- ☐ listing my achievements and ambitions

Other things:

- ☐ I am taking my medication as prescribed

It's quite a list, and we don't imagine that you are going to manage to do all of these things, or that you **need** to do all of them.

But it does show just how many things **you can do** to fight back against coronary artery disease.

If you have other medical conditions, think about how the changes you have made have improved other aspects of your health.

The changes you have made will help you stay healthier and improve many aspects of your general well being, not just your coronary artery disease. Continue to build on the improvements you have made for a happier and healthier future.

The Future

We hope that you have enjoyed using this Manual, that it has answered some of your questions and that it has helped your recovery. We also hope that the book and the CD/audio will go on being useful in the following months.

Please fill in your End of Programme Questionnaire on pages 171-174 and send it (along with your Baseline Questionnaire) to the Heart Manual Department.

Part 3

Facts and Advice to Help Your Recovery

Contents	Page
Wound healing	141
Medicines	142
Hospital tests	150
Angina	151
Other chest pain	153
Breathlessness	154
Hyperventilation	155
Palpitation	155
Anxiety about your coronary artery disease	156
Low spirits (depression)	167
Stress in relationships	170
Sleep	170
Patient Questionnaire	171
The Exercise Plan	175

Wound healing

If you have had a bypass operation your wounds will take several weeks to heal fully. The wounds can feel tender and you may experience swelling and bruising. Sensations such as tingling, numbness and itching are also common at the wound sites. You may also notice a slightly hard lumpy feeling as the new tissue forms, along with a pulling sensation around the stitches as the wound heals. Most stitches do not need to be removed as they are dissolvable.

Check your wounds daily, and call your family doctor or nurse if you notice any of the following:

- increased redness around any of the wounds
- oozing from the wounds
- increased pain
- excessive swelling or warmth of the wounds.

What you can do to help your wounds heal

- Keep your wounds clean. Prevent the build-up of bacteria by showering daily.
- Avoid bringing your wounds into direct contact with perfumed soaps, shower gels, talcum powder, deodorant sprays, perfume or creams.
- Get plenty of rest and sleep, as the healing process takes place during this time.
- Protect your wounds from injury.
- Wear loose, comfortable clothing next to your wounds, preferably made of cotton. Avoid wearing clothes made of wool or 'fluffy' materials that have loose fibres, as these can stick to your wounds and cause discomfort, as well as slowing down the healing process.
- Ensure that you have a well-balanced diet that contains protein (e.g. lean meat and fish), calcium (e.g. low fat dairy products), vitamins and minerals (e.g. fruit and vegetables).
- Do not poke, pick or scratch your wounds.
- Do not use any dressings other than those advised by the doctor or nurse.
- Do not expose your wounds to sunlight for at least 6 to 8 weeks. Then use a protective sunscreen lotion.

Breastbone healing

It takes at least 3 months for the bone to fully heal so avoid pushing down on your arms to help you get out of bed or a chair. It is also important that you avoid heavy lifting, pushing or pulling during this time.

It is not uncommon to feel as if the breast-bone is moving slightly or to hear a slight clicking sound. This will disappear as the bone settles and begins to heal. If you notice that your chest is becoming more painful or these symptoms are getting worse speak to your doctor.

It is normal to feel a little uncomfortable around your shoulders and neck. Try to sit up straight using cushions for support and remember to regularly move your arms and shoulders to help reduce stiffness. Remember to follow the advice given to you by your physiotherapist. You may have been given advice on ways to reduce pain especially when coughing or sneezing as well as breathing exercises.

Swelling

You may experience some swelling of your arms or legs particularly if you have had a blood vessel taken from this area. This is quite normal. To reduce this, it is helpful to raise the affected parts, particularly when sitting, and avoid crossing your legs.

Support stockings

These stockings are worn to promote good circulation in the legs and to prevent blood clots from developing during the first few weeks after your surgery. You may be advised to take them off last thing at night and put them back on first thing in the morning. It is very important that the stockings are put on properly. This can be difficult so ask someone to help if possible. If you are having any problems, speak to your facilitator or community nurse. You should also ensure that the stockings are kept clean, and that they do not wrinkle or roll down your leg.

Your facilitator or doctor will advise you when to stop wearing your stockings. In most cases this will be around 6 weeks.

Medicines

Like most people with coronary artery disease, you have probably been given several different kinds of pills to take.

The doctor or nurse may have explained to you what they all were and why you have to take them.

If you are like most people in the same situation, you were probably too concerned about your condition to take it all in at the time.

So we have included this section in the Manual to answer some of the most common questions you might have about your medicines.

I feel much better – so why do I need to take medicines?

Doctors prescribe medicines for coronary artery disease for two reasons:

- 1 to treat any problems caused by the disease **or**
- 2 to reduce your chances of having problems in the future.

We shall deal with each of these in turn.

1 Treating problems caused by your coronary artery disease

By 'problems' we mean:

- **symptoms** which you complain of, such as any remaining chest pains or breathlessness (more about this later)
- **signs** which doctors find when they examine you, such as fluid in your lungs or ankles, or signs which they find on your tests, such as an enlarged heart or irregular heart rhythm. We have medicines to treat many of these things.

Medicines (continued)

2 Reducing your risks for the future

In Parts 1 and 2 of the Manual we told you about the way your heart works and about the risk factors that lead to coronary artery disease. We explained what you can do to reduce your risk of future problems. But there are also many things that your doctor can do to help.

For instance:

- Heart attacks are caused by a clot forming in an artery which has coronary artery disease. One way to make this less likely to happen is to make your blood less 'sticky.' The simplest and most effective way to do this is to take an aspirin every day.
- Some tablets have been shown to reduce the risks associated with coronary artery disease. You may have been given medication to help to control risk factors like high cholesterol or high blood pressure.

So all these tablets don't mean that I'm really ill?

Not at all. Nowadays it is quite normal to be taking at least three medicines if you have coronary artery disease. **The number of pills you are given has very little to do with the amount of disease in your arteries.**

What about side-effects?

All tablets have side-effects, but most of these are not serious. For instance, almost all medicines are capable of causing rashes, which should be discussed urgently with your doctor. Any other side-effects, such as stomach upsets, headaches, etc., should also be discussed with your doctor or facilitator.

There are good reasons why you have been given each of the pills that you are taking, but if a pill makes you feel ill, your doctor may prescribe something else. There's always a choice, so don't be afraid to ask.

Can I take supplements?

Some over the counter medicines, herbal remedies, supplements and even foods like grapefruit or leafy greens can affect how certain medicines work. In some cases they can make the drug work faster, make it stronger, or alternatively not work as well as it should.

It is important to tell your doctor or pharmacist if you are taking, or wish to take any supplements. Read the information leaflet that comes with your medicine and take it as your doctor prescribes.

If I don't like one of the pills, can I stop taking it myself?

Absolutely not!

It can be dangerous to stop some pills suddenly, because the body needs time to adjust to doing without them. If you really want to stop taking them, discuss this with your doctor or facilitator. It may be possible for you to take fewer pills or lower doses, and together you can agree a plan for this.

What if I forget to take them?

Missing the odd dose probably won't cause serious problems.

If it is a pill that you take once a day, there will still be plenty of the drug in your body the next day, so just take your next dose as usual.

If it is a pill that you take two or three times a day, the next dose will be due soon and will 'top up' your levels then. If you are concerned, contact your doctor or facilitator.

The best thing is not to forget your pills in the first place. Make a list of which medicines to take at which times, and connect it with something you usually do at that time – such as your morning wash, eating lunch, getting changed after work, or going to bed. If you always take your pills at these times it will soon become a habit.

I find all the names confusing

How drug names work

Drug names **are** confusing.

The information on the next few pages may be helpful.

Any pill might be described in three different ways.

Here is an example.

a Beta-blocker

The kind of drug it is.

b Atenolol

The **name of the chemical** in the drug.

c Tenormin

The **trade name** (usually begins with a capital letter). Different companies sell the same chemical under different trade names, so one chemical may have more than one trade name.

Below are some of the most common kinds of drugs used. It isn't possible to list every drug, so if one of your pills isn't mentioned, ask your doctor what kind it is.

1 Antiplatelet agents

These medicines are the real risk reducers- probably the most important pills you can take if you have coronary artery disease. **Aspirin** is the drug most commonly used to make the platelets in the blood less 'sticky'. This reduces the likelihood of clots forming in, and then blocking diseased arteries.

A low dose of aspirin a day can markedly reduce the risk of heart attack or stroke.

We know that this risk is reduced even further if aspirin is taken alongside another type of antiplatelet agent. Examples of these additional antiplatelet agents are:

- ticagrelor
- prasugrel
- clopidogrel.

Taking aspirin **and** one of the drugs above is called dual antiplatelet therapy (DAPT). You may be given this treatment for a period after your surgery or procedure. Your doctor will decide if this is the right treatment for you and how long you should take both tablets. This will depend on any other medical conditions you may have and if you have any issues whilst taking the tablets. If you have no problems taking the low dose aspirin, this will be a lifelong treatment.

If you have asthma or problems with your stomach, taking aspirin may not be possible, or you will need to take an additional pill to protect your stomach.

A side-effect of making your blood less 'sticky' is that you may bleed more easily, but the benefits of taking these tablets are important in managing your risk. Only take the doses prescribed, more is not better!

2 Beta-blockers

These medicines are used to help control angina and support a weakened heart.

They block the effects of adrenaline, slow down the heart rate and make the heart muscle work less hard.

The chemical names of beta-blockers usually end in '-olol.'

Examples of beta-blockers include:

- atenolol
- bisoprolol
- metoprolol.

When you first start on a beta-blocker your dose may be slowly increased by your doctor. This is normal. You should tell your facilitator or doctor if you have any side-effects.

Side-effects can include tiredness, cold fingers and toes, wheeziness (these drugs should normally be avoided in patients with a history of asthma), nightmares, and difficulty with erections.

These side-effects can often be dealt with by reducing the dose or changing the particular type of beta-blocker after consulting with your doctor.

You should not stop beta-blockers suddenly, as this may cause severe angina. Make sure that you always have a good supply.

3 Statins

These medicines are yet another risk reducer!

You know that high cholesterol is a risk factor for coronary artery disease. For many patients, lowering their cholesterol level will reduce the risks associated with coronary artery disease.

Statins are given routinely to all patients who are known to have coronary artery disease, even if your cholesterol level is within the normal range. Statins work by reducing the amount of cholesterol made in your liver, but there is increasing evidence that they also have some anti-inflammatory effects which benefit the heart.

Examples of statins include:

- atorvastatin
- rosuvastatin
- simvastatin.

These medicines work very quickly and lower your risk within weeks especially when combined with changes to your diet.

Side-effects can include tiredness, muscle pains, stomach upsets and headache.

Blood tests will be done to monitor the effect of the drug on your cholesterol level.

Occasionally these medicines can cause inflammation in the muscles or liver. This may be found on blood tests. You should report any new muscle pain, tenderness or weakness immediately to your doctor. If this happens your doctor may change you on to a different statin, ask you to take a lower dose, a dose every other day, or to stop the tablet for a time and then restart it. These steps often help overcome any initial side effects.

You may be advised to avoid eating grapefruit or drinking grapefruit juice as it can interact with some statins, increasing the chances of some side-effects.

Sometimes making diet changes and taking a statin is still not enough to manage your cholesterol levels. Your doctor may prescribe other medicines to take alongside your statin or refer you to a specialist clinic.



4 Angiotensin-converting enzyme (ACE) inhibitors and angiotensin-receptor blockers (ARB's)

These medicines relax the blood vessels and can lower **your** blood pressure.

They also improve the pumping action of the heart.

This may help breathlessness or tiredness.

The chemical names of ACE inhibitors usually end in '-pril.'

Examples of ACE inhibitors include:

- **lisinopril**
- **ramipril.**

When you first start on an ACE inhibitor your dose will be slowly increased by your doctor to allow your body to get used to the effects of the drug, this is normal practice. Your doctor will check your blood pressure and take blood tests. Remember to tell your doctor if you have any side-effects, such as, dizziness when standing, persistent cough or taste disturbance.

ACE inhibitors work better for some people than others. If they don't work for you, or you are unable to take an ACE inhibitor, you may be prescribed an angiotensin-receptor blocker (ARB) instead. Like ACE inhibitors, ARB's relax the blood vessels and lower blood pressure.

The chemical names of angiotensin-receptor blockers usually end in '-sartan'.

Examples of ARB's include:

- **losartan**
- **candesartan.**

5 Nitrates

These medicines are used to control angina.

Nitrates work mainly by relaxing the veins that take blood into the heart. This has the effect of reducing the amount of blood returning to the heart, so that the heart has less work to do. These medicines may also relax the arteries, particularly the coronary arteries, improving the blood supply to your heart muscle.

If you had angina it is likely that you had a short acting glyceryl trinitrate (GTN) spray which you squirted under your tongue if you had chest pain. If so, you should also have been given advice on how to use it.

The usual advice is to use the spray and wait for 5 minutes. If you still have the pain, repeat the dose 5 minutes later.

If however you pain is unbearable or getting worse, or if you still have pain 5 minutes after the second dose, dial 999 or your local emergency number and ask for an ambulance. Tell them it is a suspected heart attack. If in doubt, read page 41 in Part 2 of the Manual.

If you had angina, you may have been advised to take your GTN to prevent episodes occurring. Read page 151-153 for more information on angina.

There are also other forms of GTN such as short or long acting tablets.

Examples of these are:

- **glycerol trinitrate**
- **isosorbide mononitrate.**

Side-effects can include dizziness when standing up and headaches. The headaches tend to disappear after the first couple of weeks of use.

Some GTN tablets only last 8 weeks once the pack is opened, so if you use GTN tablets keep track of when you opened them and make sure you get a new supply after 8 weeks.

6 Calcium-channel blockers

You may be given a calcium-channel blocker if you are not able to take a beta-blocker.

These medicines help treat angina and high blood pressure, and occasionally are given for palpitations or heart rhythm irregularities.

Like nitrates, they relax the blood vessels, but they also act more effectively on the arteries leading from the heart. They make it easier to pump blood out to the body, reducing the blood pressure. They can also relax the coronary arteries to increase blood flow to the heart muscle. In some cases they relax the heart muscle itself and slow the heart rate.

Examples of calcium-channel blockers include:

- **amlodipine**
- **diltiazem**
- **verapamil.**

Side-effects can include flushing of the face, ankle swelling, nausea, constipation and headache.

7 Diuretics

These medicines are used to remove excess water which can build up in the body and cause ankle swelling or breathlessness.

Some of them can be used to treat high blood pressure.

They work on your kidneys to make you pass more urine.

Examples of diuretics include:

- **furosemide**
- **bumetanide**
- **indapamide**
- **eplerenone.**

Side-effects can include losing too much fluid, leading to feelings of tiredness or dizziness when you stand up. Other effects can include muscle cramps, headache or feeling sick.

It is important to try to reduce your intake of salty foods such as bacon, cheese, crisps, peanuts and Bovril or Marmite, which can encourage the build-up of fluid.

As diuretics make your kidneys work harder your doctor will want to monitor them by occasionally taking some blood tests.

8 Other kinds of drug

Some patients require medicines to treat irregularities in the heart rhythm. These are known as **anti-arrhythmics**.

Your cardiologist will advise you how long you may need to take these.

Examples of anti-arrhythmics include:

- **digoxin**
- **amiodarone**
- **verapamil.**

These medicines are all very different to each other, so be sure to read the information leaflets to check for side-effects.

Anticoagulants work to 'thin the blood' meaning it stops the blood from clotting so easily. Warfarin is one of these drugs and has been used for many years. There are also newer types of anticoagulants called Direct Acting Oral Anticoagulants (DOAC's). These include:

- **rivaroxaban**
- **apixaban**
- **dabigatran**
- **edoxaban.**

These may be needed as a short-term treatment after your operation to reduce the risk of clots forming in the veins in your legs or inside your heart, particularly if you have had

problems with your heart rhythm soon after your operation.

If you have continuing problems with your heart rhythm or with the valves inside your heart, you may need anticoagulant treatment for life.

As anticoagulants 'thin the blood' you may bruise easily and bleed more from cuts. If this gets worse over time you should inform your doctor. The side-effects from anticoagulants can be serious but there are important benefits to taking them and you should speak to your doctor if you are worried.

You should avoid aspirin if you are taking an anticoagulant, unless advised by your doctor. You should also check with your pharmacist or doctor before taking over the counter medicines, herbal remedies, or supplements as some can interfere with how the anticoagulant works.

If you are taking warfarin, you will need regular blood tests to check you are on the right dose. There are some foods and drinks that you should avoid, and certain diet changes may affect the dose you need. Your doctor or nurse will discuss these with you before starting on warfarin.

There isn't space in this Manual to list all the hundreds of medicines that are available, and new ones are being developed all the time. If you have been given pills to take that are not listed on these pages, ask your doctor or facilitator what they are for.

Here is a summary of the main medicines and what they can be used for. It may be helpful to fill in the table with the names of the medicines you take, and how often **you** take them. Anything you are unsure about you can ask your doctor or facilitator.

	Reduce risk	Treat risk factors	Treat angina	Improve heart as a pump	Treat the heart rhythm	I take this type of medicine (tick)	Drug name	How many times a day?	Questions/ Comment
Antiplatelet agents	✓								
Beta-blockers	✓		✓	✓	✓				
Statins	✓	Cholesterol							
ACE inhibitors or ARB's	✓	Blood pressure		✓					
Nitrates			✓						
Calcium-channel blockers		Blood pressure	✓		✓				
Diuretics		Blood pressure		✓					
Anti-arrhythmics					✓				
Anticoagulants	✓								
Other									
Example: Antiplatelet						✓	Aspirin (low dose)	Once in the morning	Take it with or after breakfast (not on an empty stomach)with a glass of water

- Prescription medicines come with an information sheet which contains important information about dosage, possible side-effects, what to do if you miss a dose, etc. Always be sure to read the information sheet right through when you start taking the pills.
- If you are not sure how to get your medicines when you need a new supply ask your GP, they may be able to set you up with a repeat prescription. This means you don't need to see the doctor every time you need more of your prescribed medicines.
- Make sure you always have enough medication and order your new prescriptions in time, so you won't run out before the new supply arrives. Sometimes this can take a few weeks, so try to plan ahead.
- Try to get into a routine of what you take and when. This means you will be less likely to miss a dose.
- If you find it hard to open your medicine bottles or get the tablets out of the blister-packs, let your facilitator know. They can discuss what options are available to help with this.
- If you have any questions or concerns about side effects, or taking any of your tablets, talk to your pharmacist, doctor or facilitator, there may be other options or medicines that will help with the side-effects.

You can write down any other questions or concerns you may have about your medicines in the space below:

Name of medicine:

Questions:

Name of medicine:

Questions:

Name of medicine:

Questions:

Hospital tests

This section tells you about some of the routine tests you might expect when you are in hospital or when you go back for a follow-up appointment after your procedure.

Exercise tolerance test (ETT or treadmill test)

The **exercise tolerance test (ETT)** is usually carried out using a motor-driven treadmill and a special ECG machine. You walk on the treadmill while your heart rate, blood pressure and ECG are being recorded. Every 3 minutes the treadmill speeds up or gets steeper so that you have to work harder and your heart rate and blood pressure increase.

The test is used to find out how well you are recovering physically and whether you have angina. The information that it provides can also help the rehabilitation team to advise you on further exercise.

Echocardiogram (Echo)

This is just like the ultrasound test that is performed on pregnant women. It is completely painless. Sound waves are used to produce a video of the heart beating. This allows the doctor to see how well your heart is recovering and pumping.

Coronary angiogram (catheter study)

This test is used to find regions where the coronary arteries are diseased. A fine tube called a **catheter** is threaded up from the artery in your arm or leg into your heart. A special dye which shows up on X-rays is injected through the tube into the coronary arteries and filmed using an X-ray camera placed over your chest. A local anaesthetic is used to numb the area.

The X-ray pictures help your doctor to decide what, if anything, needs to be done to improve the blood supply to your heart muscle. The test takes about half an hour to complete, and you may have to lie flat in bed for a few hours.

Angina

What causes angina?

Coronary artery disease can cause several symptoms such as, shortness of breath, dizziness, changes in the way your heart beats and angina.

Angina is a pain in the chest that occurs when the coronary arteries are not getting enough blood to part of the heart muscle to allow it to work properly.

It's a type of **cramp** in the heart muscle. **Just like a cramp in any muscle, it passes with time and doesn't do any lasting damage.**

Some people experience angina as a tightness or discomfort in the chest rather than a pain. Occasionally it may be felt in the arm, the back or the jaw.

In Parts 1 and 2 of the Manual we explained how **coronary artery disease** causes your arteries to become narrowed by the build-up of cholesterol and scar tissue on the artery walls.

Your heart is just like any other muscle. The harder it works, the more oxygen it needs. Oxygen is carried in the bloodstream to the heart via your coronary arteries.

If your arteries are healthy, they can provide your heart with as much oxygen as it needs. They can stretch to let more blood and therefore oxygen through when your heart is working harder than normal. When you have coronary artery disease the arteries can't do this as efficiently.

Remember what we said on page 52 (Part 2) about rubbish being washed down a river. The rubbish builds up at certain points. Where the river gets narrower, the water still flows through the narrow parts. But as the river gets narrower still, it may reach a stage where it begins to dry up downstream.

It's the same inside your arteries. They can get to the point where so much rubbish gets stuck at a narrow part that your narrowed arteries might not be able to supply all of the blood and oxygen that your heart needs.

You may not have noticed this at first, but there may have been times when your heart was working harder than usual for instance, when you were running for a bus or walking up a hill and this brought on your angina.

Occasionally an artery may become completely blocked so that the blood can't get through at all. Over time the heart may try to deal with a blockage by creating new or using other smaller blood vessels. This is known as collateral circulation.

Angina (continued)

What brings on angina?

Most of the time angina is brought on by exercise and eases with rest. Sometimes you may have noticed your symptoms when you were not exercising. Strong emotions may have brought it on, too.

Imagine that you went out for a walk and your angina came on sooner or worse than you expected. So you thought that your angina might be getting worse. This may have caused you to tense up and feel anxious. Emotions can make your body release adrenaline into your blood (p156-166). This will have made your heart beat faster. The faster your heart was

beating, the harder it had to work. The harder it worked, the more oxygen it needed, so the more angina you felt. The more angina you felt, the more anxious you may have become. The more anxious you become, the more adrenaline you produced ... get the idea?

It's easy to see how a vicious circle can be set up.

Other things could have brought your angina on too, such as cold and windy days or eating a heavy meal.

When it is cold your body tries to keep warm by making the blood vessels in your arms and legs

smaller. This squeezes blood to the centre of your body making your heart work harder. When it is windy it takes more effort to walk and it can be difficult to breathe so your heart has to work harder again.

Angina can also be worse after a heavy meal, when your heart has to pump a lot of extra blood to your stomach to help your digestion. This means two things. First, your heart has to work harder for an hour or so after the meal. Second, there's less blood for your muscles or your brain (that's why you get sleepy!).

Will my angina go away?

Your procedure improved the blood supply to your heart, and in most cases this will have got rid of your angina straight away. However, sometimes it takes a bit longer for the heart to 'fine tune' the new blood supply. In

a few weeks or months, as you recover and follow the advice in the Manual, the angina may well disappear completely. Remember what you know about the heart's amazing powers of recovery. Your body

can often help itself by finding a different way around any remaining narrowed sections of artery, re-routing the blood supply to the heart.

Angina (continued)

What can I do about angina?

In an earlier section we said that angina can be brought on by exercise and strong emotions. So you might think that in order to get rid of it you should stop taking exercise and avoid getting upset.

Wrong! That would be easier said than done – and anyway it doesn't work. If you stop exercising, you become less fit and it takes more effort to do even simple things. Avoiding exercise is the surest way to make angina come on more readily.

Avoiding stress is virtually impossible, too – and remember that some stress and excitement is a normal part of life.

So, to keep angina in check:

- Keep up your exercise/activity plan to gradually build up fitness. If you get angina when you are exercising, slow down or

stop. As you get fitter, this should happen less often.

- Remind yourself that angina is not like a heart attack. It is simply your heart's way of saying 'Hold on a minute – I need to catch up.'
- If the pain comes on when you are upset, check that you are breathing correctly, using your stomach muscles and not your chest. Do your relaxation and check that you aren't tense. Wait for the pain to ease off, and congratulate yourself on dealing with it calmly.
- If it is cold and windy wrap up warm and try to shelter from the wind. Some people find wearing a scarf gently over their mouths also helps if they get pain when breathing in cold air.
- If your angina comes on after a heavy meal try eating

smaller portions and make sure that you do not exercise too soon after eating.

- You may have been given a GTN spray to use. This can help to stop angina more quickly. Carry the spray with you at all times, and use it as well as doing the things listed above. If you are about to do something which you know will bring on angina, use your spray before starting.

Avoiding exercise and stress doesn't work. However, managing exercise and stress can be part of your treatment, and you can take some precautions.

But if your pain returns or is severe don't ignore it follow the advice on page 41 (Part 2 of the Manual) and speak to your doctor.

Other chest pains

Chest pains often don't mean angina – or that you are having a heart attack. Other things can cause chest pain, too.

Think back to before you were told you had coronary artery disease.

If you are like most people, you would have felt unusual pains in your chest – and other parts of

your body – from time to time. People who haven't got a heart condition usually put this sort of thing down to 'twinges' or cramps or creaking joints.

The pains are often caused by pain messages which travel up special nerves to the brain.

Sometimes these nerves produce 'false alarms' which

may be our body's way of testing the system.

It's not surprising if your body decides to test the system more often after a heart procedure. And it's not surprising if you notice pains and twinges more than you did before. Don't worry about these pains – they will settle down in time.

Other chest pains (continued)

Chest pains

Some chest pains have physical causes, like the pain from your chest wound if you had a bypass operation or your body adjusting after having an angioplasty or stent. Others are caused or made worse by worry and anxiety.

Anxiety can make these pains worse in all kinds of ways:

- It causes muscle tension, which can lead to pain.
- It makes you much more aware of twinges and false alarms.

- It can make you breathe too fast, causing chest pain (see 'Hyperventilation' section on page 155).

The wrong kind of breathing can cause chest pain if you breathe with your chest muscles. Look back at page 36 of Part 2 of the Manual, and remember that there are special breathing exercises on the relaxation CD/audio to help with this.

So with all these possible causes of chest pains, it's easy to see how you can get into a vicious circle.

You feel a pain in your chest – you worry about it – the worry makes it feel worse – you worry more – you get dizzy and sweaty – and so on. It isn't surprising that some people end up back in hospital!

If you are getting pains in your chest, try to remember all of these things. Make sure that you are doing your relaxation and breathing properly. This will help.

But if it's severe pain and none of this advice helps, don't ignore it – speak to your doctor or follow the advice on page 41, Part 2 of the Manual.

Breathlessness

Breathlessness

Normally we don't think about our breathing. We just do it. But if we run for a bus, we breathe faster and we notice it – we feel breathless.

This can be because you are actually breathing faster, or simply because you are noticing your breathing more.

Coronary artery disease can affect the way that your heart works as a pump, and this may make you breathless. Usually this is treated with medicines. You may also feel breathless especially after a bypass

operation if you have developed a chest infection or if you have a little fluid which has gathered around your lungs. If you have been asked to monitor your breathlessness or take tablets to treat these things, it is important to speak to your doctor if your breathlessness isn't improving or if you think it is getting worse.

If the problem continues, your cardiologist may request further tests or offer other forms of treatment.

But by far the commonest reason for feeling breathless is simply the fact that **you are noticing your breathing more**. Increased awareness of breathing should settle within a few weeks after your procedure but, just like chest pain, it is made worse by anxiety. So work on your relaxation and breathing.

One particular type of breathlessness requires a little more explanation. It is called **hyperventilation**.

Hyperventilation

Hyperventilation

On the relaxation CD/audio we told you about the best way to breathe. Hyperventilation is one type of wrong breathing, and it can cause chest pain. Often people who are rushed into hospital with a suspected heart attack turn out to be hyperventilating or having a panic attack (see Laura May story on page 161).

Some people hyperventilate all the time and some only do so when they are under stress.

Very often the person who is hyperventilating is not aware of the problem, but those close to them may notice that they often sigh, or that their breathing often seems very fast, or that they sometimes stop breathing and then take a deep breath.

Another sign is that their chest is often heaving up and down. Because hyperventilation is difficult for you to spot in yourself, ask someone you

know well if they notice you doing any of these things. Breathing like this makes you feel nervous, frustrated or twitchy. This is because it triggers the release of adrenaline into the blood stream.

Here are other symptoms:

- feelings of light-headedness
- the sensation that things around you are far away or unreal
- a feeling of tingling or numbness in the mouth, fingers or tongue
- a feeling that you can't get enough air
- wobbly legs
- a feeling that you must 'get away' from something.

What can I do about it?

- Practise the breathing exercises on the CD/audio at least once a day.
- Try to check your breathing

often during the day, and correct it if you are breathing with your chest.

- Get your partner to remind you to breathe with your stomach.
- When you are making an effort – such as opening a door, getting out of a chair, opening a tight jar, etc. – **try not to hold your breath**, as this often sets it off.

If you keep practising it will get better – but it may take quite a long time, because you've probably been hyperventilating for years. If, after several months of practising, the problem is no better, or if you are having real difficulties getting your breathing right, ask your GP or facilitator to refer you to a clinical psychologist.

Palpitation

Palpitation

This is another symptom which may simply be due to being more aware of your body. Palpitation means being aware of your heartbeat. Just as with breathlessness, we usually only notice it if our heartbeat is faster than normal.

You may notice it more for some weeks after your procedure. Some medicines may affect your pulse and make you more aware of it, too.

Palpitation can be made worse by anxiety, because adrenaline increases the heart rate.

And of course your heart rate will increase when you exercise. If you are unfit, it may remain higher for a time even after you stop.

Sometimes palpitation can be due to a problem. You might be aware of your heart missing a beat or putting in an extra one.

It is quite common to have an unusual heart beat for a short period of time after a procedure, especially a bypass operation. You may have even been given medication to help control this.

If your heart has periods when it starts racing or the beat doesn't seem to be regular, let your doctor or nurse know. You might need more tests or a special type of medicine called an anti-arrhythmic to control the problem.

Anxiety about your coronary artery disease

It's normal to worry if you have had a procedure. Who wouldn't worry in the same situation? Getting more information about worry or anxiety can help many people to handle it better. That is what this section is about.

Anxiety can cause some or all of these physical feelings:

- dry mouth
- 'butterflies' or tightness in the stomach
- a cold sweat
- rapid pulse
- palpitations
- rapid breathing
- cold, clammy hands and feet
- tense, sore muscles
- a sensation of being far away or distant from what's going on
- feeling faint
- strange pains
- tingling in the feet and hands
- ... and many other feelings.

Anxiety has these cognitive or 'mental' effects:

- difficulty in concentrating
- difficulty with remembering
- difficulty with decision making
- racing thoughts
- problems with sleeping
- lack of self-confidence
- feeling that your personality has changed
- feeling that you are going mad.

Anxiety can cause these kinds of behaviour:

- not listening to what people are saying
- feeling unsatisfied
- restlessness
- fidgeting
- bad temper
- flaring up at little things
- losing your sense of humour
- becoming overcritical or sensitive
- becoming socially isolated.

We all get like this when we are worried, scared or under stress. The primitive emotion responsible is fear.

What's happening in my body?

When we are in danger or under stress, our bodies automatically produce chemicals to help us to cope.

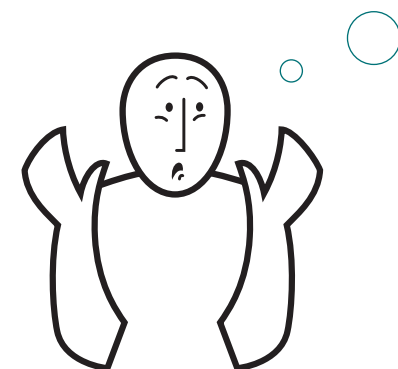
The main chemical that we produce is **adrenaline**. In a real emergency it can help to save our lives.

Imagine that you are crossing the road. Suddenly a lorry comes hurtling towards you and this thought flashes through your mind:

This lorry isn't going to stop. I'll be killed!

The thought would probably be so quick that you wouldn't notice it. It triggers the production of adrenaline, and your adrenaline level goes up. The adrenaline gets you ready in an instant to take avoiding action.

It isn't only frightening events that make us produce adrenaline. Frightening words, thoughts or memories trigger it, too. This is the same for everyone.



Anxiety (continued)

Adrenaline gets us ready to fight or run when we are in danger. This is what it does:

Effects on the mind

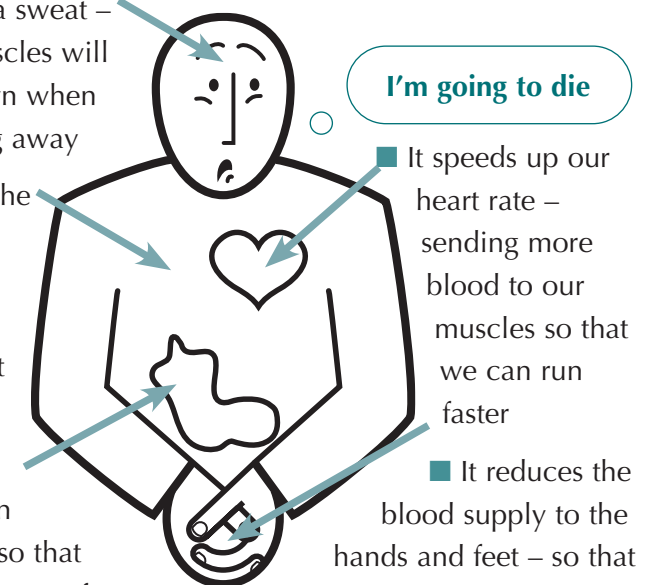
- It makes us concentrate on the danger and ignore other less important things.
- It makes our thoughts race – so that we can look for quick ways to escape.
- It makes us 'think the worst.'

All animals have this 'fight or flight' mechanism. It's what helps them to survive.

Another name for it is **fear**. If we didn't feel fear we would be a danger to ourselves and to others.

Effects on the body

- It produces a sweat – so that the muscles will be cooled down when we are running away
- It increases the breathing rate, sending more oxygen to our muscles so that we can run faster
- It speeds up our heart rate – sending more blood to our muscles so that we can run faster
- It reduces the blood supply to the hands and feet – so that there is more for the main muscles
- It shuts down the stomach – so that there is more energy for running



The problems start when we make the adrenaline – but there's no one to fight and we can't run away ('fight or flight syndrome')

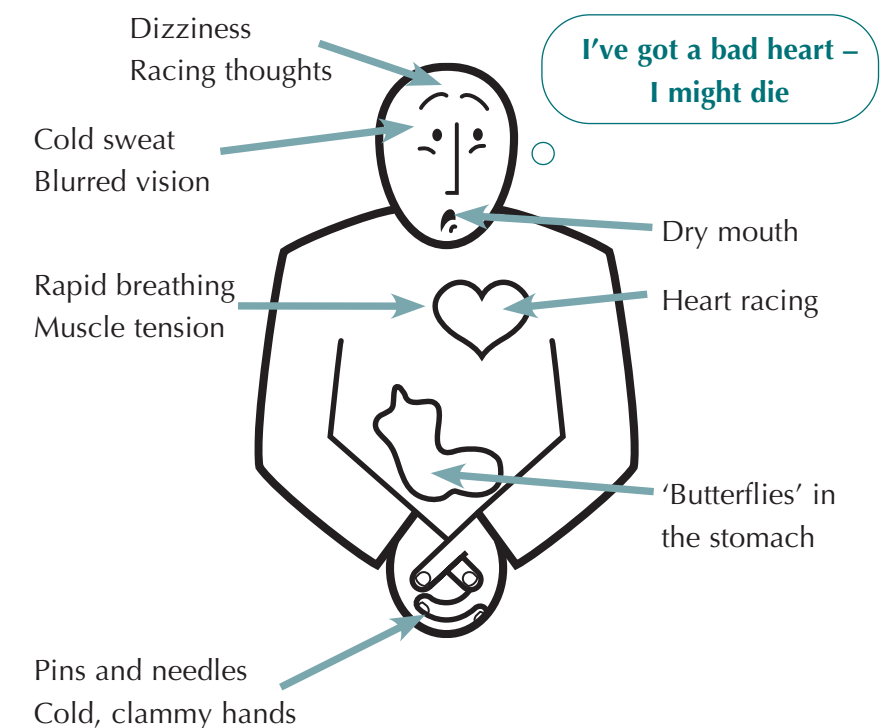
A frightening thought, image or memory can produce adrenaline in the same way as a frightening event.

But this time there's no one to fight and nowhere to run.

So what happens to the extra adrenaline? It doesn't get used up, and it causes unpleasant physical and mental symptoms.

If we were running we wouldn't notice these symptoms – but as we are probably just going about our normal routine, we do notice them.

The racing heart, cold sweat, dizziness and other symptoms make us feel scared and ill.



The anxiety story so far

Any worrying thought, memory or event triggers the production of adrenaline.

There is nothing wrong with this. It's part of nature's way of helping us to cope with danger so that we can live to fight another day.

If we are in *real* danger we do run or fight. We are too busy to notice the effects that adrenaline has on our body.

But what if a worrying thought happens when you are sitting in a chair, or trying to get to sleep, or shopping? It can cause very unpleasant feelings.

- It makes your thoughts race, so that you can't relax, or it may make it difficult for you to get to sleep.
- It increases your breathing rate making you feel dizzy or faint and far away.
- It speeds up your heart rate – you may notice that your heart is racing or pounding.
- It shuts down your stomach, giving you a feeling of 'butterflies' or a knotted stomach or a sinking feeling,
- It produces an unpleasant cold sweat – it may make your forehead feel clammy.
- It shuts down the blood supply to your hands and feet – you may notice that they feel cold and clammy.

Why is this common after a procedure?

Do you remember the Stress Thermometer? It shows how you feel at different levels of adrenaline.

The more you relax the lower it gets.

- Panic X
- Very scared X
- Scared X
- Worried X
- Bad-tempered X
- Tense X
- Excited X
- Alert ✓
- Wide awake ✓
- Calm ✓
- Sleepy ✓
- Very relaxed ✓



These sensations will not harm you. They are just unpleasant. They will go away as you recover and realise that you are getting better.

As you know, having a procedure can be very frightening.

This means that your adrenaline level is already raised.

Any worrying thought or event can be enough to put your Stress Thermometer level up from alert to worried – or higher.

However, relaxation lowers the level.

The vicious circle

If you notice these bodily feelings, this can lead to thoughts like:

- 'There is something wrong with my heart again.'
- 'I'm not getting better – I'm getting worse.'
- 'I feel faint – I may be going to pass out.'
- 'If I don't get to sleep soon I'll be ill.'
- 'I'm having a heart attack.'

Of course these thoughts will

produce more adrenaline, so the bodily feelings get worse.

If we notice this, we worry even more – this makes the feelings worse, and so on.

This won't hurt you, but it's very unpleasant, especially if you don't know why it's happening!

Occasionally, in some people, this vicious circle can go on quickly getting worse and worse until the adrenaline builds up to a very high level and they have what is called a **panic attack**.

In a full panic attack most people are convinced that they are dying.

However, a panic attack is not dangerous and it is usually over within 10–30 minutes.

This is because after a while the body runs out of adrenaline.

The person who is having the panic attack begins to notice that they are getting better.

They think:

'It's stopping – I'm safe.'

'It wasn't a heart attack after all.'

'I'm going to be all right.'

'I'm feeling better all the time.'

Thoughts like these switch off the production of adrenaline.

They slow down the pulse, calm the breathing, reduce the heart rate and stop the stomach churning – and the person feels better and better.

Many people experience sudden increases in the amount of adrenaline in their blood.

It isn't just something that happens to people who have coronary artery disease.

It causes a feeling of being unwell, or feeling scared or angry.

This can be because a picture, a memory or even a smell has brought a worrying thought into

the person's mind.

After a procedure your adrenaline level is already quite high. These thoughts produce even more adrenaline.

The extra adrenaline raises the temperature on your Stress Thermometer, making you feel worried, tense or bad-tempered.

The feelings may quickly pass, or you may end up feeling tense and unhappy most of the time.

The medical name for this is **anxiety**.

It is not dangerous – just unpleasant.

It is common in people under any kind of stress – such as divorce, serious money problems, unemployment, etc.

It is very common in people who have had a serious illness.

We know from research that these thoughts almost always stop after a while.

Until they do, you may find that thoughts about your heart and your illness are always in your mind.

'I feel exactly like that, but I've never noticed myself having sudden worrying thoughts.'

Very often these thoughts can pass through your mind so quickly that you don't notice them. What you do notice are the effects that the thoughts set off.

You can tell that this has happened if you suddenly feel:

- worried
- tense
- scared
- bad-tempered
- 'butterflies' in your stomach
- cold sweats
- weak
- tired
- sad
- breathless
- any of the other symptoms of anxiety.

These very quick thoughts, of which we are hardly aware, are called **negative automatic thoughts**.

Learning the true facts about heart disease can help to fight these thoughts.

On the next three pages are some examples of how anxiety has affected other people.

Ian Gray's story

Anxiety can cause bad temper.

After his bypass operation, Ian Gray had returned to his work as storeman at a large garage.

His job involved fetching parts from the storage racks and then filling out paperwork. His chest had healed up well after the operation, and he didn't find light lifting a problem. Sometimes he had very little to do. At other times he would have a rush of work, with people queuing up waiting for him to get the parts they needed.

He found that he was getting very bad-tempered with the other people at work.

Three or four times he had suddenly felt very ill and had been sent home.

After a while he began to hate his work, and he dreaded going in each day.

When he wasn't at work he felt fine, except when he thought of having to go to work.

Soon his dislike of work meant that he was taking a lot of time off.

He said that he hated all the people there and he wanted to give up work.

He felt that he wasn't strong enough for it any more.

He went to see his doctor who, after doing a number of tests, told him that there was no longer anything physically wrong with him.

When the effects of anxiety were explained to Ian, he realised that he became bad-tempered mainly when there was a long queue of people all waiting for him to serve them. Using a simple method that he had been shown, he was able to discover the automatic thoughts that occurred each time the queue built up.

These thoughts were:

'Having all these people waiting is putting me under stress.'

'Stress is bad for the heart.'

'It could kill me.'

It is not surprising that Ian was getting annoyed.

Although he hadn't been aware of it, he was really thinking:

'These people could kill me!'

Knowing what his automatic thoughts were didn't cure Ian. He still found himself getting angry and anxious whenever a queue built up.

But he did get better, and this is how:

■ First he learned relaxation, and he practised relaxing every evening.

■ This brought down the temperature on his Stress Thermometer. It made him feel a bit calmer during the day.

■ While he was at work, if he noticed any symptoms, such as 'butterflies' in his stomach or a cold sweat, he used the 'rapid relaxation' technique.

■ This relaxed his muscles, which helped to reduce the tension.

■ To stop the negative thoughts (about stress damaging his heart), he repeated to himself in his mind:

'I'm quite relaxed, which is good for my heart.'

'The more I relax, the better I feel.'

After a few weeks of this he felt OK again, and he was soon working happily with no more worries about stress.

Laura May's story

If you get tired or breathless or feel your heart beating fast, it can make you think that there is something wrong with your heart. This can cause anxiety, which makes you feel much worse.

A good example of this is what happened to Laura May.

She had a close and loving family.

When she developed angina, they were all very shocked and scared, especially her husband, John.

John took over a lot of the things that Laura had once done, including the shopping.

He often told her that shopping at the supermarket would be too much for her.

But she was actually quite fit (probably fitter than John, who took very little exercise).

After a few months she had an angioplasty (stent or balloon), which got rid of the angina.

One day, Laura decided that she would do the week's shopping to show John that she could manage again.

She went to the local

supermarket, which was very crowded.

She began to feel rather short of air and could feel a slight sweat on her forehead.

After she had filled her trolley, she found that the queue at the checkout was very long.

As she stood in the queue, she began to feel as if things around her were miles away, like looking down the wrong end of a telescope.

This feeling frightened her. She also noticed that her heartbeat was very fast.

She thought that this might mean she was having a heart attack.

By now she was desperate to get out of the shop, but she didn't want to draw attention to herself by just walking away and leaving her trolley.

She hardly heard a thing that the woman at the checkout said to her, and she can't remember how she got out of the shop.

All she could think was:

'I'm having a heart attack.'

Somehow she managed to phone her husband at work.

He drove her to the hospital. After a brief chat about the symptoms Laura had felt, the doctor was able to tell her that she had had a panic attack.

She told Laura about the physical symptoms of anxiety.

Laura realised that she had been feeling rather worried before she went into the supermarket.

Most people feel a little anxious in crowded places, and this had also raised her adrenaline level.

Once she thought about it, she remembered that shopping in that particular store had often made her feel uncomfortably hot and short of air because it was overheated.

Because she was anxious about her heart, she had mistaken these feelings.

This had automatically made her think:

'Maybe John is right – I'm not strong enough to do this.'

'Maybe it will be bad for my heart.'

'Maybe I'll have a heart attack and die.'

This had produced even more adrenaline – enough to make her feel sick.

Then she had noticed her heart racing. This made her think:

'There is something wrong with my heart. I can feel it!'

With these thoughts going round and round in her head, she produced more and more adrenaline, which eventually caused a panic attack.

The same kind of thing often happens if a person feels tired all of a sudden, or notices a strange pain or new sensation.

Anxiety (continued)

John Robertson's story

A common experience is to suddenly feel anxious or tired, or unwilling when you think about having to do something.

This is how anxiety affected John Robertson.

Before he got angina and had a bypass operation, John had been a keen gardener.

While he was in hospital and during the first few weeks after he returned home, he was looking forward to getting back to gardening because the garden was in a mess.

After a couple of months he was driving again and fit enough to walk 5 miles every day.

But he still did nothing about the garden.

Although he often intended to start, whenever he looked at the mess it had become he felt weak and tired, and put it off.

Eventually even hearing the garden mentioned, or catching a glimpse of it out of the window, made his stomach churn and his heart speed up.

After reading about anxiety, he realised that every time he thought about the garden, these negative automatic thoughts were going through his mind:

'I ought to do the garden.'

'It might damage my chest.'

'It might be too much for my heart and I might die.'

In other words, any thought of the garden made him think of injury and dying.

John's wife often remarked that he seemed to have lost interest and that he should start gardening again.

These negative thoughts were triggering off the adrenaline that was making him feel unwell and tired – and making him avoid the gardening.

Once he realised what the thoughts were, he was able to notice this happening and he corrected the negative thoughts by thinking calming thoughts instead. He thought:

'I ought to do the garden.'

'So long as I take it carefully and stop if I get too tired, it won't hurt me.'

'In fact it may make me fitter, which would be good.'

Thinking about it in this new way reduced the anxiety, and he was soon gardening again. After a while he was enjoying it more than ever.

Anxiety: summing up

It is normal to feel anxious after a heart operation or any other very frightening experience.

This anxiety usually gets less as the memories fade.

It is caused by a raised adrenaline level. This is normal after a frightening experience.

It can affect us in two ways:

- 1** The raised adrenaline level may cause a general feeling of tension that is present all the time. It may make sleeping difficult. We may feel generally restless and unable to concentrate on anything. We may find that our anger flares up quickly, or we may seem cut off and become distant. It may make us unwilling to go out, take exercise, etc.
- 2** We may feel all right most of the time, but get sudden feelings of 'butterflies' in the stomach, weakness, or a feeling of dread as if something terrible is about to happen. This is usually because of an automatic negative thought or memory, generally about the danger of having a bad heart. Our body can't tell the difference between a real danger (such as a lorry heading straight towards us) and a fear about our heart. It just does its job and produces more adrenaline. This can result in our suddenly feeling ill, peculiar or scared. Although this is unpleasant, it is not dangerous. Of course we may experience both 1 and 2.

Anxiety (continued)

Some solutions

Sudden anxiety

As soon as you notice yourself getting anxious, think about the statements below. They are called the Six Rules for Coping with Anxiety.

You may like to cut out the box below and carry it with you.

Cut along dotted line 

Cut along dotted line 

Six Rules for Coping with Anxiety

- 1** It is not my heart or mind that is going wrong.
- 2** The feelings that I have are the effects of adrenaline.
- 3** These feelings are not dangerous – just unpleasant.
- 4** Notice what is happening in my body right now – not what I fear may happen.
- 5** Remember that when I stop adding to my anxiety with negative thoughts, it starts to fade by itself.
- 6** Wait, and give the fear time to pass. Do not fight it or run away from it. Just accept it.

Doing any of the rapid relaxation techniques that work for you will help the feelings to go more quickly.

Anxiety (continued)

See if you can identify the negative thought that has produced the sudden rise in adrenaline.

These thoughts happen very quickly.

The best thing to do is to try and remember what you were thinking about just before the feelings appeared.

For example, in John Robertson's case the thought was 'I ought to do the garden.'

At first he didn't realise why this thought should be a problem, so he used the 'So Why' test.

The 'So Why' test.

'I should do the garden.'



'So why' should that worry me?

'It might damage my chest.'



'So why' should that worry me?

'It might be too much for my heart.'



'So why' should that bother me?

'I might die.'

This is also called the downward-arrow technique because each 'So why' gets you further down towards the negative automatic thought.

Here is another example from Ian Gray.

'There is a queue building up.'



'So why' should that worry you?

'It puts me under pressure.'



'So why' should that worry you?

'Pressure is bad for the heart.'



'So why' should that worry you?

'I might have a heart attack.'



'So why' should that worry you?

'I might die.'

Use the 'So Why' test

Step 1

'What was I thinking about, looking at or remembering just before I started feeling like this?'

Step 2



'So why' should that worry me?

Answer:



'So why' should that worry me?

Answer:



'So why' should that worry me?

Anxiety (continued)

Identifying negative thoughts is a great help because they are usually wrong.

For example, doing the garden was very unlikely to kill John Robertson.

As soon as he realised that this was what he was really worried about, he could see that it was unreasonable, and the garden no longer bothered him.

The only part of the negative thought that John was aware of was 'I ought to do the garden.'

Until he did the 'So Why' test he couldn't understand why this should upset him.

Once he knew why it did, he could deal with the problem.

What can you do if the fear is a reasonable one?

One answer is to test it out.

Take John Robertson as an example.

It is just possible that his thought 'It might be too much for my heart' was correct.

The sensible thing to do is to approach the feared thing a bit at a time.

Step 1: Work out what you think is definitely safe.

John could do the garden for 10 minutes and then see if he was OK.

Step 2: Increase it by an amount that you think is safe.

The next day he could have worked for 20 minutes, and so on, until he felt completely confident that no harm could come to him.

You can use this method to overcome almost any fear.

Suppose that, like a lot of people, you are worried about going out for the first time after your operation.

Work out how far you could go without feeling worried – for example, to the end of the garden path. Just walk that far a few times until you are sure that it will do you no harm.

Choose the next target – for example, walking to the end of the street. Do that a few times, and so on.

The reason why this works is because after a while you stop believing the negative automatic thought.

You stop believing it because you have proof that the thing that you are worrying about isn't going to happen.

Are you good at advising other people? If you are, **treat your own worries as if they were someone else's.**

We can usually give other people good advice about their problems. It's not so easy when it comes to our own, because we can't 'get outside the problem.'

Pretending that it is someone else who is worrying about the problem often helps us to see the situation in a more calm and relaxed way.

Do a rapid relaxation exercise.

This will help to lower the adrenaline level more quickly than just waiting for it to go down.

Replace the negative automatic thoughts with positive ones.

This is called positive self-talk, and it is another technique taught to sportsmen and women to help them to play better and with more confidence.

Positive thoughts have the opposite effect to negative ones. They switch off the production of adrenaline.

Every time you notice a negative thought, replace it with a positive thought.

Negative thought
'Gardening may damage my heart.'

Positive thought

'Rubbish! It will probably strengthen it.'

Other positive thoughts include statements like:

'I'm doing fine.'

'This is easy.'

'I'm much fitter than I was just a few weeks ago.'

You can find a number of positive statements on page 40 in Part 2 of the Manual.

The positive statements will work much better if you **think** about each one as you read it.

Another technique used by sportsmen and women is **positive rehearsal**.

They get into a relaxed frame of mind and then imagine their next game.

They imagine themselves playing brilliantly, scoring goals or winning each set, or whatever they want to achieve. Doing this can actually help it to happen!

Suppose you become aware that you are worried about what the first day back at work is going to be like.

Or perhaps your next hospital appointment.

Do your relaxation and then imagine the whole thing in your mind. Imagine the waiting room, being called in by the nurse, shaking hands with the doctor, and asking all of the questions that you would like to ask but usually forget or can't bring yourself to mention.

That's just one example. You can use this for anything that you are going to have to do and which is worrying you – your first walk down to the shops, your first time driving again, and so on.

Imagine this happening as it would if everything was perfect – a sunny day, a nice doctor, the perfect drive, and so on.

Or picture yourself in a couple of months' time – relaxed and cheerful in any situation you can imagine.

What can you do about constant tension and restlessness?

You can do all of the things that we have just mentioned – plus regular sessions of relaxation twice a day for 20 minutes.

Remember to keep checking for tension during the day.

- Are you frowning?
- Is your neck tense?
- Are your shoulders tense?
- Is your breathing as smooth and regular as it could be?
- Are your hands clenched into fists?

You may get fed up doing this, but the more often you undo this tension, the more relaxed you will eventually become. Don't despair – keep at it!

You can set reminders to do it – for example, each time the adverts come on the TV, or each time you make a cup of tea, or each time the phone rings.

Exercise is good

It helps to burn up the spare adrenaline. If you feel tense, go for a short walk or do some of your home exercises.

Low spirits after a heart procedure (depression)

Low spirits are common after a heart procedure. These feelings usually disappear as time passes.

They can range from the occasional feeling of sadness or tearfulness to absolute despair.

All of these symptoms are common:

- sadness
- tearfulness
- poor appetite
- early waking
- loss of interest in things that were once enjoyable
- no energy for doing things
- loss of interest in appearance.

These symptoms are not dangerous, just very unpleasant. And it is worth remembering

that the physical effects of the operation itself can make people feel tired, have sleep problems, etc.

If these feelings become very severe they are called **depression**.

One of the most unpleasant features of depression is what it does to the way a person thinks.

A depressed person can usually only see the bad side of things.

Even things that in the past would have seemed enjoyable can seem flat and dull.

Negative automatic thoughts

If you have read the section on anxiety, you will already know about negative automatic thoughts.

We all have automatic thoughts all the time.

They are the kind of everyday thoughts that pass through our mind constantly without us paying much attention to them.

For example, the phone rings and you think 'I must answer the phone.'

If someone had asked you what you were thinking about when the phone rang, you might well say 'Nothing.'

But your body doesn't just act on its own – your brain had to recognise the sound of the phone ringing and tell your body what to do.

When the phone rings, you automatically think 'I must answer the phone.'

This thought is so quick and so obvious that you don't notice it has happened.

By now you may be wondering what all of this has got to do with feeling miserable after a heart procedure.

The telephone ringing sets off an automatic thought – 'I must answer the phone.'

In the same way, after a heart operation anything that reminds you of it or of your angina can set off **negative** automatic thoughts such as:

- 'I'm finished.'
- 'I'm no use any more.'
- 'My life is as good as over.'
- 'I'll just get worse again.'
- 'I'll soon be dead.'

These kinds of thoughts make you feel sad.

In fact they would make anyone feel sad – if they believed them.

The trouble with automatic thoughts is that they happen so fast we don't get time to consider whether they are true or not.

We just automatically believe them.

The result is often a sudden feeling of sadness that we can't explain.

Here's an example of how this can happen.

Pam Clark suddenly felt tearful when she was talking to her sister on the phone.

At first she couldn't work out why. Then she remembered that her sister had asked her about

(continued overleaf)

Depression (continued)

her last holiday, and this thought had flitted through her mind:

'I might never have another holiday.'

Sometimes it may be the sight of something, or a memory, or even a smell, that can trigger off these thoughts.

If you can work out what the thought was, this can give you a chance to put it right.

For example, Pam was able to remind herself of the fact that heart procedures work extremely well for most people.

She also knew that she had reduced her risk factors because she had lost weight, learned to relax, given up smoking and taken up regular exercise.

She was able to remind herself that she had every chance of making a good recovery.

Once she thought about it logically, using the true facts rather than the automatic thoughts, she felt better.

You can find some useful ways of identifying negative automatic thoughts on pages 163–164 of this Manual.

If you can spot them and think about them more logically, the sadness usually goes.

What to do about sudden sadness

The first thing to do is to get it in perspective.

These sudden feelings of sadness happen to most people, but they almost always fade away as time passes.

If they only happen now and then, and most of the time you feel OK, it is best just to accept them as part of getting better.

This is a stage that most people have to go through, and the more you experience it now the sooner it will be over.

Remind yourself that although sadness is unpleasant, it is not dangerous.

The thoughts are almost always wrong.

We know from research that there are several different kinds of wrong thoughts.

Crystal ball gazing

This means guessing what will happen. We all do this all the time, but when we are sad or afraid we do it in a negative way. We always imagine the worst happening.

For example:

'I'll probably get angina again.'

'I'll have to give up work.'

'This will last for ever.'

Unless you really are a fortune teller and can see into the future, there is no way that you can guess whether these things are true and, as we have already said, the chances are that none of them will happen.

Exaggerating

The slightest setback or difficulty may lead to thoughts of complete disaster. For example, you may feel tired and think:

'I'll never be able to work again.'

Feeling guilty

You may blame yourself for what has happened:

'All this is my own fault.'

'If I were a better person none of this would have happened.'

Ignoring the positive things that happen

For example, Jean, a relative, phones up to ask how you are, and instead of thinking how nice it is to hear from her, you may think:

'She's only phoning because she thinks I'm at death's door.'

All of these thoughts have one thing in common. They are biased towards the negative.

Before your coronary artery disease started to get you down, you would have seen everything differently.

In a few weeks or months from now you will see things more clearly again.

Depression (continued)

Negative thinking + lack of activity = MISERY

'What's the point?' and 'I can't be bothered' are thoughts that often occur to people who are depressed.

This isn't because of laziness. It's the way that depression makes them feel about the world.

The result is that they probably do very little.

They may even avoid the things that used to give them pleasure.

Or, if they do take part in an enjoyable activity, negative thoughts often ruin it for them.

Thoughts like:

'I shouldn't be enjoying myself like this.'

'I'm no fun any more.'

And so on.

Cutting down on activities in this way means that they don't experience the normal excitement that we all need in order to stay happy and alert.

This leads to them feeling even worse.

Feeling worse means that they do less, and so it goes on in a vicious circle.

It is important to break this circle by keeping busy with enjoyable activities.

It can take a lot of willpower, because you may not feel like doing things. However, once you start to do them you will often find that you enjoy them more than you thought you would.

It will be much easier to stick to this if you make a plan.

Activity scheduling

Every day should contain some enjoyable activity.

MON	Go to library
TUE	Buy new shirt
WED	Visit Bob Smith
THUR	Walk round park
FRI	Meet friends for lunch
SAT	Go shopping
SUN	Walk round harbour.

■ Make a list of the days of the week and make sure that there is some activity for each day.

■ For example, going out, seeing friends, buying yourself a small gift or taking exercise will all help.

■ You may not enjoy these activities as much as you would normally, but you will feel better than you would if you just sat around.

If these feelings get very bad, and after a month or so they seem to be getting much worse, you should tell your doctor. They may be able to prescribe a pill that can help, or if you would like to know more about how to control negative thinking they may be able to arrange for you to see a clinical psychologist.



Stress in relationships



Bad relationships at home are a common source of stress.

Of course most couples have rows from time to time, but constant bad feelings can make life a misery.

One of the commonest reasons for problems between partners is resentment.

This often builds up because one partner feels that they are being taken for granted, or that they have to do everything around the house, or that their own enjoyment of life is being sacrificed to the other person or to the family.

Sometimes it's just that the stresses and strains of a busy life, work and worry lead to a lack of communication and shared pleasures.

Sleep

People sometimes have trouble sleeping after a procedure. Problems are more common after a bypass operation than angioplasty (stent or balloon). The trouble can last up to four weeks although most people find that their sleep improves a week after surgery.

- Problems can start because of:
- The effects of the anaesthetic
 - Wound discomfort
 - Changes in sleep patterns after your hospital stay
 - Worry, anxiety or depression

These things are rarely one person's fault. It is important to be honest and realise that you probably have faults as well.

Talking things over can be a help. Getting away on holiday together and resolving to make a new start can also help. This may be a good time to start to put the romance back in the relationship. If things have got to the point where trying to talk just leads to more rows, you should seriously consider seeing a counsellor or a psychologist.

There is a great advantage in having an experienced person to help you sort things out. Many thousands of couples have improved their relationships and their lives in this way.

Most people find it unpleasant to think about discussing their

private affairs with a stranger. But people who have done it often say that they wish they had done so years ago!

Some people don't like the idea because they think that the counsellor will side with their partner.

This will not happen, because the people who do this work are trained to see both sides of the situation, and they will be more interested in suggesting practical ways of overcoming the problem than in raking over the past.

The relationship counselling service is listed in the phone book under the name 'Relate' or 'Relationships Scotland'.

If there isn't a service in your area, your doctor can make an appointment for you to see a psychologist.

Tips to improve sleep:

- Avoid prolonged napping during the day unless you are still able to sleep at night, in which case it is better to limit your naps to less than 40 minutes.
- Avoid caffeine (coffee, tea, caffeinated fizzy drinks)
- Take regular exercise as well as rest
- Try to get out into the fresh air.
- Avoid exercising too close too bed time (leave a good hour)
- Practise relaxation

- If worrying at night keeps you awake it helps to write your worries down to deal with the next day
- Keep your bedroom only for sleep and sex – read or watch television in another room
- Use a wind down routine before bedtime, no exciting computer games or thriller films

Sleep problems can be a sign of anxiety and depression, please look at pages 156-169 and discuss any concerns with your facilitator or GP.

Baseline Patient Questionnaire



SCAN ME

Please complete this questionnaire at the start of your Heart Manual Programme. You will be asked to complete a similar questionnaire (next page) at the end of the programme. This will allow you to compare where you are now, and at the end of the 6 weeks using the Heart Manual. This information is also very useful to us and we will ask you to return both questionnaires to the Heart Manual Department at the end of the programme.

You can access the questionnaire by responding below, or scan the QR code above, or use the online link: <https://app.onlinesurveys.jisc.ac.uk/s/nhslothiansurveys/revasc-baseline>

☐ I consent to take part in this survey and understand how my information will be used – please tick.

Please circle the most appropriate rating for each statement:

	Strongly Disagree	Disagree	Unsure	Agree	Strongly Agree
I understand my condition	1	2	3	4	5
I understand the procedure and why I needed it	1	2	3	4	5
I understand my medication	1	2	3	4	5
I understand my blood pressure	1	2	3	4	5
I understand how to manage stress and anxiety	1	2	3	4	5
I understand how to manage low spirits	1	2	3	4	5
I understand how to deal with sexual problems	1	2	3	4	5
I understand how to deal with sleep problems	1	2	3	4	5
I understand how to deal with a wound	1	2	3	4	5
I understand how to manage pain	1	2	3	4	5
I set goals	1	2	3	4	5
I pace my activities	1	2	3	4	5
I feel motivated to change my lifestyle	1	2	3	4	5
I am confident in managing my other conditions	1	2	3	4	5
I can influence my recovery	1	2	3	4	5

In the last 6 months:

	Never	Rarely	Occasionally	Frequently	Daily
I exercise	1	2	3	4	5
I practise relaxation	1	2	3	4	5
I practise correct breathing	1	2	3	4	5
I smoke	1	2	3	4	5
I eat a healthy diet	1	2	3	4	5

What are your main concerns/worries right now?

.....

.....

End of Programme Patient Questionnaire

Please fill in this final questionnaire once you have completed your cardiac rehabilitation programme. Then, take a moment to look back and compare your answers in Section 1, to the answers you gave in the ‘Baseline Patient Questionnaire’ at the start of the programme. You may be surprised by the differences!

Providing this feedback to us is invaluable. We will use your feedback for audit and evaluation purposes which will help manual users in the future. These audit findings may be published. No one will be able to identify you from them.

Please return the completed ‘Baseline’ and ‘End of Programme’ questionnaires together to the address given at the bottom of the survey.

You can access the questionnaire by responding below, or scan the QR code above, or use the online link: <https://app.onlinesurveys.jisc.ac.uk/s/nhslthiansurveys/revasc-end-of-programme>

☐ I consent to take part in this survey and understand how my information will be used – please tick.

Please circle the most appropriate answer for each question

What procedure did you have?	Angioplasty/Stent	☐	Bypass	☐
Did you read the whole of the manual?				
Did anyone you live with read the manual? (if applicable)	Yes		No	
How many times did your Heart Manual facilitator either visit you at home or speak to you on the telephone?				
Home visit	Telephone	Other (please state)		

Please circle the most appropriate rating for each statement:					
	Strongly Disagree	Disagree	Unsure	Agree	Strongly Agree
I understand my condition	1	2	3	4	5
I understand the procedure and why I needed it	1	2	3	4	5
I understand my medication	1	2	3	4	5
I understand my blood pressure	1	2	3	4	5
I understand how to manage stress and anxiety	1	2	3	4	5
I understand how to manage low spirits	1	2	3	4	5
I understand how to deal with sexual problems	1	2	3	4	5
I understand how to deal with sleep problems	1	2	3	4	5
I understand how to deal with a wound	1	2	3	4	5
I understand how to manage pain	1	2	3	4	5



SCAN ME

Cut along dotted line

Please circle the most appropriate rating for each statement:

	Strongly Disagree	Disagree	Unsure	Agree	Strongly Agree
I set goals	1	2	3	4	5
I pace my activities	1	2	3	4	5
I feel motivated to change my lifestyle	1	2	3	4	5
I am confident in managing my other conditions	1	2	3	4	5
I can influence my recovery	1	2	3	4	5

In the last 6 months:

	Never	Rarely	Occasionally	Frequently	Daily
I exercise	1	2	3	4	5
I practise relaxation	1	2	3	4	5
I practise correct breathing	1	2	3	4	5
I smoke	1	2	3	4	5
I eat a healthy diet	1	2	3	4	5

Please circle the most appropriate rating for each statement:

	Strongly Disagree	Disagree	Unsure	Agree	Strongly Agree
The manual is very clear about when to call a doctor	1	2	3	4	5
Overall, the information in the manual was easy to understand	1	2	3	4	5
The manual was easy to use	1	2	3	4	5
Overall the manual was useful to me	1	2	3	4	5
Overall the exercise plan/video was useful to me	1	2	3	4	5
Overall the relaxation audio was useful to me	1	2	3	4	5
Overall the question & answer audio was useful to me	1	2	3	4	5
This manual has helped me to eat a healthier diet	1	2	3	4	5
This manual has helped me to recover	1	2	3	4	5
This manual has helped me to increase my daily activity levels	1	2	3	4	5
Having contact with my facilitator helped me	1	2	3	4	5
I would have liked more contact with my facilitator	1	2	3	4	5
Using the Heart Manual has helped me to manage my condition	1	2	3	4	5

How has the Heart Manual helped you with other conditions? If applicable (e.g. diabetes, arthritis, etc.)

What do you think has changed in your day to day life since completing the programme?

Is there anything that you would like to see added to the Heart Manual?

What sections do you think have been most useful?

Any other comments about the Heart Manual?

Has the Heart Manual helped you to address your main concerns? (See baseline questionnaire)

Thank you. Please return both questionnaires to:

The Heart Manual Department

Astley Ainslie Hospital, Grange Loan, Edinburgh EH9 2HL

Contact telephone number: 0131 537 9127/9137

Email: heart.manual@nhs.scot • Website: www.theheartmanual.com

If you feel that your experience could assist the development of the Heart Manual in the future please contact us using the details above.

Cut along dotted line

The Home Exercise Plan

The home exercise plan is described on the next five pages. Please read all five pages carefully all the way through. You can also find a demonstration video of the home exercises online at:

www.heartmanual.scot.nhs.uk/index.php/download/resources
or scan this QR code.



As you have learned, to make you fitter, exercise should make you breathe faster and feel warmer. This type of exercise (e.g. your walking plan) is called 'aerobic' exercise. However, there are other aspects of fitness which are also important. These are strength, balance and flexibility. The exercises on the following pages will help you develop all these aspects of fitness.

There are four sets of exercises:

- 1. Stretches** – these should be done after every exercise session, to encourage flexibility.
- 2. Leg Strength Exercises** – these will help build strength in your legs. You should try to do these twice per week, but not two days in a row (e.g. Monday & Wednesday).
- 3. Arm Strength Exercises** – these will help build strength in your arms. You should try to do these twice per week, but not two days in a row (e.g. Tuesday & Thursday).
- 4. Aerobic Exercises** – these can be used to supplement your walking, or as an alternative to walking on days that you are unable to get out.

The combination of these exercises (particularly the stretches and leg exercises) can also help to improve balance.

The exercises are suitable for people who are starting at a low level after a bypass operation or for those who did not participate in regular exercise before their procedure, but will be beneficial for everyone, particularly in the early stages of recovery. If you have had an angioplasty and were quite fit before the procedure, you may wish to think about starting at a higher level or doing another form of exercise.

When doing the exercises:

- Do 'warm-up' before you start your exercises, and gradually 'cool down' once you have finished. This could be walking around the room for 1-2 minutes if you are doing 5 minutes of exercise, but should be increased as you gradually build up the amount of time you are exercising.
- Do the stretches after you have completed your exercise as instructed below.
- Build up in stages - only do what you are able to do comfortably, when it becomes 'easy' for at least 2 days in a row gradually increase them.
- If you have a short break when doing any of the exercises (e.g. to have a drink of water) keep your feet gently moving to maintain circulation.
- Fill in your exercise record after you have done your exercises.

Instructions on how to do the exercises are given on each page. Read the instructions carefully before starting your exercise plan. Your facilitator should do the exercises with you before you get started. This will help to ensure that you are doing the exercises safely.

The stretches

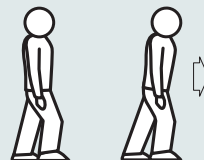
It is important that you do these stretches after every exercise session, to encourage flexibility. Stand to do the stretches, do them smoothly and don't bounce or jerk. If you need extra balance, hold on to something firm which will not move like the edge of the kitchen work top or the back of the sofa.

Hold each stretch for 5 seconds to start with, aiming to build up each stretch to about 20 seconds, repeat with the other leg or arm.

Starting position: Standing

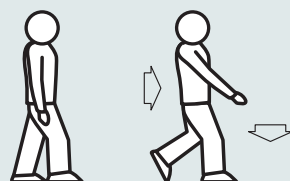
1. Calf stretch

Place one foot in front of the other, with your front leg bent and your back leg straight. Gently move your weight forward over the front leg, keeping the heel of the back leg on the ground. You should feel a stretch in your calf muscle.



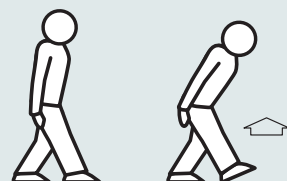
2. *Quadriceps stretch

Stand with one foot in front of the other. Move your body weight mainly to the front foot, keeping it flat on the floor, and raise your heel on the back foot. Widen your stance or hold on to something for support if you feel unbalanced. Bend both knees slightly and tuck your tail bone under. You should feel the stretch in the front of your back thigh.



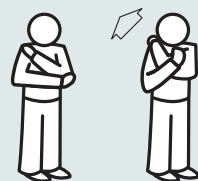
3. Hamstring stretch

Place one foot in front of the other with your front leg straight, your back leg bent and your weight on the back leg. Rest your hands on the back of your thigh for support. Pull up the toes on the front leg so that you are resting on your heel. You should feel the stretch on the back of your front leg.



4. Shoulder stretch

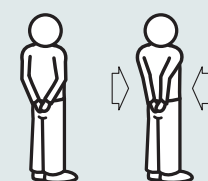
Place one arm across your chest to the opposite shoulder. Gently stretch the upper arm with the other hand. You should feel the stretch across your shoulder and upper back.



5. Chest stretch

Place both hands at the small of your back. Keeping your shoulders down and back, gently bring your elbows towards one another.

Modification: if you have had a bypass do this very gently at first, and avoid this completely if you are experiencing problems with healing of your chest scar.



The stretches can be done safely by most people although the *Quadriceps stretch should be avoided in the first week after angioplasty if the upper leg has been used. All stretches should be avoided until wounds have healed. It is however important to continue your exercise and activity. This may mean that you have to be active for shorter periods of time and concentrate on cooling down rather than stretching.

Strengthening exercises

These exercises will help to build up the strength in your arms and legs. For all the exercises:

- try to do them at a slow rhythmical pace (e.g. 3 secs outward movement, 3 secs to return to starting position).
- aim to do two sets of about 10-15 repetitions of each exercise (this applies to each limb if the exercise is done one limb at a time).
- try to keep your body upright (avoid leaning forward, or to the side) and maintain good posture throughout. Keep your breathing steady and avoid holding your breath.

If you had a bypass, **and** are having problems with the healing of your scars, please speak to your facilitator for advice on which of these exercises will be suitable for you.

Leg strengthening exercises

1. Knee bends

Starting position: stand sideways next to a table or kitchen worktop with one hand resting on it for balance.

Exercise: bend both knees slowly as far as is comfortable, then return to the starting position. Keep your knees soft and avoiding locking them.



2. Side leg lifts

Starting position: stand facing a table or kitchen worktop with one hand resting on it for balance.

Exercise: slowly lift one leg out to the side, as far as is comfortable. Keep the knee straight, toes pointing forward and both hips facing forwards. Slowly lower it to the starting position.



3. Leg Extension

Starting position: stand facing a table or kitchen worktop with your hands resting on it for balance

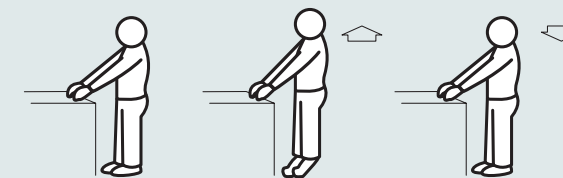
Exercise: slowly lift one leg behind you, as far as is comfortable, keeping the knee straight, then slowly return it to the starting position.



4. Heel Raises

Starting position: stand with feet slightly apart, facing a table or kitchen worktop with your hands resting on it for balance.

Exercise: slowly raise both heels off the floor, until you are on your toes. Slowly lower your heels back to the floor.



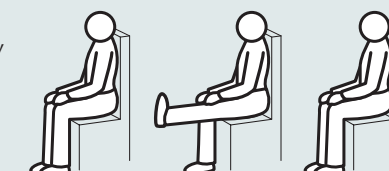
Modification:

If you are struggling to do the leg exercises in standing, you can strengthen your thigh muscles in sitting.

Knee Extension

Starting position: sitting with good posture, arms relaxed. Ideally use a chair which allows you to sit with both feet flat on the floor and supports both thighs, almost to the back of the knees.

Exercise: slowly raise one foot off the floor until the leg is straight, then bend your toes on that foot up towards your body. Slowly lower the foot and place it flat on the floor and repeat with the other leg.



Arm strengthening exercises

Remember:

- these should be done at a slow rhythmical pace.
- aim to do two sets of about 10-15 repetitions of each exercise (this applies to each limb).
- try to keep your body upright. Keep your breathing steady and avoid holding your breath.

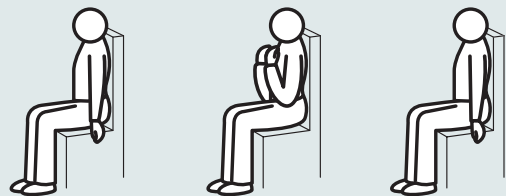
If you have had a bypass, **and** are having problems with the healing of your scars, please speak to your facilitator for advice on which of these exercises will be suitable for you.

If you have had an angioplasty and have tried these arm exercises for a week without any problems, you could try adding in some light resistance (e.g. holding small (330ml) bottles of water). Increase the resistance gradually by using **slightly** heavier objects. If you have had a bypass, the stage at which you can add in some resistance will depend on the healing of your scars, but will typically be from week six onwards. Speak to your facilitator for further guidance on this.

Bicep curls

Starting position: standing or sitting straight with good posture, arms by sides, turned outwards with palms facing forwards.

Exercise: keeping your upper arms by your sides, slowly bend both arms at the elbow and bring your hands up towards your shoulders. Slowly return your hands back to the starting position.



Upright row

Starting position: standing or sitting straight with good posture, arms in front of the body, with palms resting on thighs.

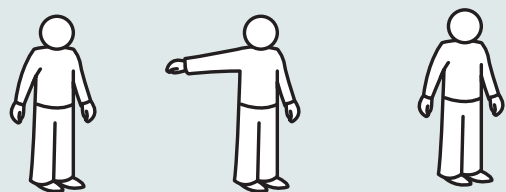
Exercise: slowly bring both hands up to your chest, with elbows out to the side as if pulling up a zip, and then return them to the starting position.



Lateral raises

Starting position: standing or sitting straight with good posture, arms by sides.

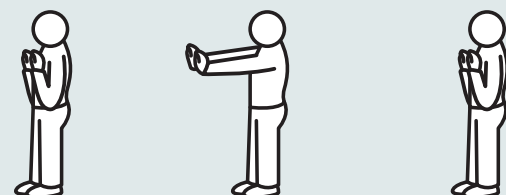
Exercise: slowly lift one arm up to the side, as far as is comfortable, but not higher than shoulder height. Slowly return your arm to the starting position.



Chest press

Starting position: standing or sitting straight with good posture, arms in front of the body, elbows bent with hands in front of your shoulders (palms facing forwards).

Exercise: slowly reach both arms forwards, until they are straight, then return them to the starting position.



Aerobic Exercises

You can do these to supplement your walking, or instead of walking on the days that you are unable to get out. You can also use them to break up periods of inactivity. Aim to do these exercises at your walking pace – breathing faster, feeling warmer, and try to do these exercises for the total time that you would walk for (e.g. if you have been doing an 8-minute walk, aim to build up 8 minutes over the four exercises). The speed at which you do these exercises will affect how difficult you find them. Try to keep up a steady pace. As you progress you may find that you want to make the exercises harder by speeding up.

Aerobic exercises

1. Marching on the spot

Starting position: standing.

Exercise: lift each knee in turn to a height that is comfortable. To start with, do this for a count of 30 seconds, gradually building up 10 seconds at a time as you find it getting easier.

Progression: bring your knees higher and bring in an arm swing.

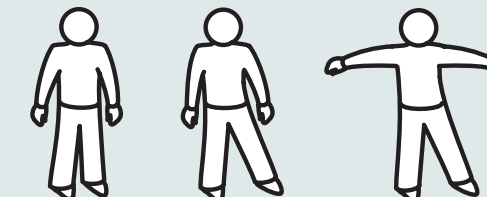


2. Side taps

Starting position: standing, feet slightly apart, your knees soft.

Exercise: Step to the side with one foot and then bring it back in. Repeat with the other leg.

Progression: raise both arms whilst stepping your feet to each side, keep your arms below shoulder level.



3. Heel digs

Starting position: standing.

Exercise: start with your feet slightly apart and your knees soft. Step one foot forward with toes pulled up in a heel dig and then bring it back. Repeat this with your other leg.

Progression: add in a bicep curl by bringing both hands up to the shoulders whilst doing each heel dig.

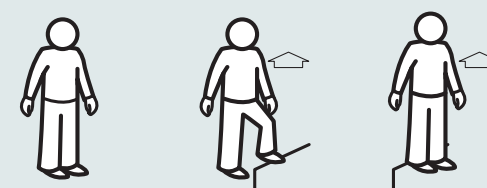


4. Step ups

Starting position: stand in front of a small step, with your feet hip distance apart. You can use the bottom stair or your front or back doorstep.

Exercise: tap one foot on to the step, and return to the starting position. Repeat with the other leg.

Progression: step up with one foot and then the other so that both feet are on the step – make sure that your whole foot is on the step – and then return to the starting position.



Modifications:

- If you find it difficult to balance or would like some extra support you may find it helpful to hold on to a firm surface which will not move, such as the edge of the kitchen worktop or the back of the sofa. Try using the hand rail while doing the 'step ups'. With time you may find that you do not need as much support.
- All of these exercises can be done while sitting. If you choose to do this, it is best to sit on a firm chair which will not move. Try to use one with a hard back for extra support, such as a dining room chair. With time you may progress to doing them holding on to a firm surface.
- If you are doing the exercises while sitting you may want to add in the arm exercises from the 'side tap' and from the 'heel dig' exercises.
- Some people find it easier to march around the room rather than 'march on the spot'. Make sure there are no obstacles in the way before you start.

Remember if you have a set back or stop exercising for a while speak to your doctor or facilitator. You can also find more advice on p108.

The Heart Manual

Revascularisation Edition



No one other than a registered and regulated healthcare professional is qualified to choose which patients are suitable for this Manual.

If you have a heart condition and have been prescribed the Manual for your own use, please do not pass it on to anyone else. If you have a heart condition and have been given the Manual by anyone other than a registered and regulated healthcare professional, please do not use the Manual.

No guarantee can be given that following this Manual programme will benefit any particular individual. All the advice and guidance contained in the Manual is based on medical opinion and knowledge current at the date of publication, but no guarantee can be given that statements made in the Manual will not be contradicted by subsequent opinion or knowledge.